Correlated random effects: `iidkd`

This model is available for dimensions $k = 2$, to 24. We describe in detail the case for $k = 3$ as other ones are similar. This model do the same as models `iid2d`, `iid3d`, `iid4d`, `iid5d`, but uses a different and more efficient parameterisation.

Parametrization

The $(k = 3)$-dimensional Normal-Wishard model is used if one want to define three vectors of "random effects", u and v and w, say, for which (u_i, v_i, w_i) are iid bivariate Normals

$$\begin{pmatrix} u_i \\ v_i \\ w_i \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \mathbf{W}^{-1}\right)$$

where the covariance matrix $\mathbf{W}^{-1}$ is parameterised as $\mathbf{W} = \mathbf{L}\mathbf{L}^T$, where

$$\mathbf{L} = \begin{pmatrix} \exp(\theta_1) & & \\ \theta_4 & \exp(\theta_2) & \\ \theta_5 & \theta_6 & \exp(\theta_3) \end{pmatrix} \tag{1}$$

and $\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6$ can take any value. The number of hyperparameters are $k(k+1)/2$, which is 3, 6, 10, 15, 21, 28, 36, 45, 55, ..., for $k = 2, 3, 4, 5, 6, 7, 8, 9, 10, \ldots$.

For these models the precision matrix $\mathbf{W}$ is Wishart distributed

$$\mathbf{W} \sim \text{Wishart}_k(r, \mathbf{R}^{-1}),$$

with density

$$\pi(\mathbf{W}) = c^{-1}|\mathbf{W}|^{(r-(k+1))/2} \exp\left\{-\frac{1}{2}\text{Trace}(\mathbf{W}\mathbf{R})\right\}, \quad r > k+1$$

and

$$c = 2^{(rk)/2}|\mathbf{R}|^{-r/2}\pi^{(k(k-1))/4}\prod_{j=1}^{k}\Gamma((r+1-j)/2).$$

Then,

$$\text{E}(\mathbf{W}) = r\mathbf{R}^{-1}, \quad \text{and} \quad \text{E}(\mathbf{W}^{-1}) = \mathbf{R}/(r-(k+1)).$$

Hyperparameters

The hyperparameters are $\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6$.

The prior-parameters are

$$(r, R_1, R_2, R_3, R_4, R_5, R_6)$$

where

$$\mathbf{R} = \begin{pmatrix} R_1 & R_4 & R_5 \\ R_4 & R_2 & R_6 \\ R_5 & R_6 & R_3 \end{pmatrix}$$

The `inla` function reports posterior distribution for the hyperparameters $\{\theta_i\}$, and the conversion into interpretable quantities can be done using simulation as described below.

The prior for θ is **fixed** to be `wishartkd`, and number of prior parameters required are $1 + k(k+1)/2$. By default the prior-parameters are

$$(r = 100, \underbrace{1, \ldots, 1}_{k \text{ times}}, 0, \ldots, 0)$$

Specification

The model `iidkd` is specified as

```
y ~ f(i, model="iidkd", order=3, n = <length>) + ...
```

where order $= k = 3$, and the `iidkd` model is represented internally as one vector of length n,

$$(u_1, u_2 \ldots, u_m, v_1, v_2, \ldots, v_m, w_1, w_2, \ldots, w_m)$$

where $n = 3m$, and n is the (required) argument in `f()`.

For this model the argument `constr=TRUE` is interpreted as 3 sum-to-zero constraints

$$\sum u_i = 0, \quad \sum v_i = 0 \quad \text{and} \quad \sum w_i = 0.$$

Hyperparameter spesification and default values

(**Note:** The value "1048576" is just a code for "replace this by the default value". As the default value depends on `order`, the was the easy way out for the moment.)

doc `Gaussian random effect in dim=k with Wishart prior`

hyper

theta1

hyperid `29101`
name `theta1`
short.name `theta1`
initial `1048576`
fixed `FALSE`
prior `wishartkd`
param `30 1048576 1048576 1048576 1048576 1048576 1048576 1048576 1048576 1048576 104`
to.theta `function(x) x`
from.theta `function(x) x`

theta2

hyperid `29102`
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short.name `theta2`
initial `1048576`
fixed `FALSE`
prior `none`
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from.theta `function(x) x`

theta3

hyperid `29103`
name `theta3`
short.name `theta3`
initial `1048576`
fixed `FALSE`

prior none
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from.theta function(x) x

theta4
hyperid 29104
name theta4
short.name theta4
initial 1048576
fixed FALSE
prior none
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from.theta function(x) x

theta5
hyperid 29105
name theta5
short.name theta5
initial 1048576
fixed FALSE
prior none
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theta6
hyperid 29106
name theta6
short.name theta6
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from.theta function(x) x

theta7
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name theta7
short.name theta7
initial 1048576
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from.theta function(x) x

theta8

hyperid 29108
name theta8
short.name theta8
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theta9

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short.name theta9
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theta10

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name theta10
short.name theta10
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theta11

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short.name theta11
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theta12

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short.name theta12

initial 1048576
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theta13

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short.name theta13
initial 1048576
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theta14

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short.name theta14
initial 1048576
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theta15

hyperid 29115
name theta15
short.name theta15
initial 1048576
fixed FALSE
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theta16

hyperid 29116
name theta16
short.name theta16
initial 1048576
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from.theta `function(x) x`

theta17

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theta19

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theta20

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short.name `theta20`
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theta21

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theta22

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name theta22
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theta23

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theta24

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theta25

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theta26
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theta32

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theta33

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theta34

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theta35

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theta36

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theta37

hyperid 29137
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short.name theta37
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theta38

hyperid 29138
name theta38
short.name theta38
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theta39

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theta43

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theta65

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theta67

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theta68

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theta69

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theta75

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theta81

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theta82

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initial 1048576
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theta83

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theta84

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theta85

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theta86

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theta87

hyperid `29187`

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short.name theta87
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theta88

hyperid 29188
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initial 1048576
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theta89

hyperid 29189
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theta90

hyperid 29190
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short.name theta90
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theta91

hyperid 29191
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hyperid 29192
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theta96

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theta97

hyperid 29197
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fixed FALSE
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theta98

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short.name theta98
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theta99

hyperid 29199
name theta99
short.name theta99
initial 1048576
fixed FALSE
prior none
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from.theta function(x) x

theta100

hyperid 29200
name theta100
short.name theta100

initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta101

hyperid 29201
name theta101
short.name theta101
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta102

hyperid 29202
name theta102
short.name theta102
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta103

hyperid 29203
name theta103
short.name theta103
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta104

hyperid 29204
name theta104
short.name theta104
initial 1048576
fixed FALSE
prior none
param

to.theta `function(x) x`
from.theta `function(x) x`

theta105
hyperid `29205`
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short.name `theta105`
initial `1048576`
fixed `FALSE`
prior `none`
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from.theta `function(x) x`

theta106
hyperid `29206`
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short.name `theta106`
initial `1048576`
fixed `FALSE`
prior `none`
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from.theta `function(x) x`

theta107
hyperid `29207`
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short.name `theta107`
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fixed `FALSE`
prior `none`
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to.theta `function(x) x`
from.theta `function(x) x`

theta108
hyperid `29208`
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short.name `theta108`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta109
hyperid `29209`

name theta109
short.name theta109
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta110

hyperid 29210
name theta110
short.name theta110
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta111

hyperid 29211
name theta111
short.name theta111
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta112

hyperid 29212
name theta112
short.name theta112
initial 1048576
fixed FALSE
prior none
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to.theta function(x) x
from.theta function(x) x

theta113

hyperid 29213
name theta113
short.name theta113
initial 1048576
fixed FALSE

prior none
param
to.theta function(x) x
from.theta function(x) x

theta114
hyperid 29214
name theta114
short.name theta114
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta115
hyperid 29215
name theta115
short.name theta115
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta116
hyperid 29216
name theta116
short.name theta116
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta117
hyperid 29217
name theta117
short.name theta117
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta118

hyperid 29218
name theta118
short.name theta118
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta119

hyperid 29219
name theta119
short.name theta119
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta120

hyperid 29220
name theta120
short.name theta120
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta121

hyperid 29221
name theta121
short.name theta121
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta122

hyperid 29222
name theta122
short.name theta122

initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta123

hyperid 29223
name theta123
short.name theta123
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta124

hyperid 29224
name theta124
short.name theta124
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta125

hyperid 29225
name theta125
short.name theta125
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta126

hyperid 29226
name theta126
short.name theta126
initial 1048576
fixed FALSE
prior none
param

to.theta `function(x) x`
from.theta `function(x) x`

theta127
hyperid `29227`
name `theta127`
short.name `theta127`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta128
hyperid `29228`
name `theta128`
short.name `theta128`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta129
hyperid `29229`
name `theta129`
short.name `theta129`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta130
hyperid `29230`
name `theta130`
short.name `theta130`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta131
hyperid `29231`

name theta131
short.name theta131
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta132

hyperid 29232
name theta132
short.name theta132
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta133

hyperid 29233
name theta133
short.name theta133
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta134

hyperid 29234
name theta134
short.name theta134
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta135

hyperid 29235
name theta135
short.name theta135
initial 1048576
fixed FALSE

prior none
param
to.theta function(x) x
from.theta function(x) x

theta136
hyperid 29236
name theta136
short.name theta136
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta137
hyperid 29237
name theta137
short.name theta137
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta138
hyperid 29238
name theta138
short.name theta138
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta139
hyperid 29239
name theta139
short.name theta139
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta140

hyperid 29240
name theta140
short.name theta140
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta141

hyperid 29241
name theta141
short.name theta141
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta142

hyperid 29242
name theta142
short.name theta142
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta143

hyperid 29243
name theta143
short.name theta143
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta144

hyperid 29244
name theta144
short.name theta144

initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta145

hyperid 29245
name theta145
short.name theta145
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta146

hyperid 29246
name theta146
short.name theta146
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta147

hyperid 29247
name theta147
short.name theta147
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta148

hyperid 29248
name theta148
short.name theta148
initial 1048576
fixed FALSE
prior none
param

to.theta `function(x) x`
from.theta `function(x) x`

theta149

hyperid `29249`
name `theta149`
short.name `theta149`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta150

hyperid `29250`
name `theta150`
short.name `theta150`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta151

hyperid `29251`
name `theta151`
short.name `theta151`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta152

hyperid `29252`
name `theta152`
short.name `theta152`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta153

hyperid `29253`

name theta153
short.name theta153
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta154

hyperid 29254
name theta154
short.name theta154
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta155

hyperid 29255
name theta155
short.name theta155
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta156

hyperid 29256
name theta156
short.name theta156
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta157

hyperid 29257
name theta157
short.name theta157
initial 1048576
fixed FALSE

prior none
param
to.theta function(x) x
from.theta function(x) x

theta158
hyperid 29258
name theta158
short.name theta158
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta159
hyperid 29259
name theta159
short.name theta159
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta160
hyperid 29260
name theta160
short.name theta160
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta161
hyperid 29261
name theta161
short.name theta161
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta162

hyperid 29262
name theta162
short.name theta162
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta163

hyperid 29263
name theta163
short.name theta163
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta164

hyperid 29264
name theta164
short.name theta164
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta165

hyperid 29265
name theta165
short.name theta165
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta166

hyperid 29266
name theta166
short.name theta166

initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta167

hyperid 29267
name theta167
short.name theta167
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta168

hyperid 29268
name theta168
short.name theta168
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta169

hyperid 29269
name theta169
short.name theta169
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta170

hyperid 29270
name theta170
short.name theta170
initial 1048576
fixed FALSE
prior none
param

to.theta `function(x) x`
from.theta `function(x) x`

theta171

hyperid `29271`
name `theta171`
short.name `theta171`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta172

hyperid `29272`
name `theta172`
short.name `theta172`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta173

hyperid `29273`
name `theta173`
short.name `theta173`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta174

hyperid `29274`
name `theta174`
short.name `theta174`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta175

hyperid `29275`

name theta175
short.name theta175
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta176

hyperid 29276
name theta176
short.name theta176
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta177

hyperid 29277
name theta177
short.name theta177
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta178

hyperid 29278
name theta178
short.name theta178
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta179

hyperid 29279
name theta179
short.name theta179
initial 1048576
fixed FALSE

prior none
param
to.theta function(x) x
from.theta function(x) x

theta180
hyperid 29280
name theta180
short.name theta180
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta181
hyperid 29281
name theta181
short.name theta181
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta182
hyperid 29282
name theta182
short.name theta182
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta183
hyperid 29283
name theta183
short.name theta183
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta184

hyperid 29284
name theta184
short.name theta184
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta185

hyperid 29285
name theta185
short.name theta185
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta186

hyperid 29286
name theta186
short.name theta186
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta187

hyperid 29287
name theta187
short.name theta187
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta188

hyperid 29288
name theta188
short.name theta188

initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta189

hyperid 29289
name theta189
short.name theta189
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta190

hyperid 29290
name theta190
short.name theta190
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta191

hyperid 29291
name theta191
short.name theta191
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta192

hyperid 29292
name theta192
short.name theta192
initial 1048576
fixed FALSE
prior none
param

to.theta `function(x) x`
from.theta `function(x) x`

theta193
hyperid `29293`
name `theta193`
short.name `theta193`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta194
hyperid `29294`
name `theta194`
short.name `theta194`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta195
hyperid `29295`
name `theta195`
short.name `theta195`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta196
hyperid `29296`
name `theta196`
short.name `theta196`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta197
hyperid `29297`

name theta197
short.name theta197
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta198

hyperid 29298
name theta198
short.name theta198
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta199

hyperid 29299
name theta199
short.name theta199
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta200

hyperid 29300
name theta200
short.name theta200
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta201

hyperid 29301
name theta201
short.name theta201
initial 1048576
fixed FALSE

prior none
param
to.theta function(x) x
from.theta function(x) x

theta202

hyperid 29302
name theta202
short.name theta202
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta203

hyperid 29303
name theta203
short.name theta203
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta204

hyperid 29304
name theta204
short.name theta204
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta205

hyperid 29305
name theta205
short.name theta205
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta206

hyperid 29306
name theta206
short.name theta206
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta207

hyperid 29307
name theta207
short.name theta207
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta208

hyperid 29308
name theta208
short.name theta208
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta209

hyperid 29309
name theta209
short.name theta209
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta210

hyperid 29310
name theta210
short.name theta210

initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta211

hyperid 29311
name theta211
short.name theta211
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta212

hyperid 29312
name theta212
short.name theta212
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta213

hyperid 29313
name theta213
short.name theta213
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta214

hyperid 29314
name theta214
short.name theta214
initial 1048576
fixed FALSE
prior none
param

to.theta `function(x) x`
from.theta `function(x) x`

theta215

hyperid `29315`
name `theta215`
short.name `theta215`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta216

hyperid `29316`
name `theta216`
short.name `theta216`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta217

hyperid `29317`
name `theta217`
short.name `theta217`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta218

hyperid `29318`
name `theta218`
short.name `theta218`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta219

hyperid `29319`

name theta219
short.name theta219
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta220

hyperid 29320
name theta220
short.name theta220
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta221

hyperid 29321
name theta221
short.name theta221
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta222

hyperid 29322
name theta222
short.name theta222
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta223

hyperid 29323
name theta223
short.name theta223
initial 1048576
fixed FALSE

prior none
param
to.theta function(x) x
from.theta function(x) x

theta224
hyperid 29324
name theta224
short.name theta224
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta225
hyperid 29325
name theta225
short.name theta225
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta226
hyperid 29326
name theta226
short.name theta226
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta227
hyperid 29327
name theta227
short.name theta227
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta228

hyperid 29328
name theta228
short.name theta228
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta229

hyperid 29329
name theta229
short.name theta229
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta230

hyperid 29330
name theta230
short.name theta230
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta231

hyperid 29331
name theta231
short.name theta231
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta232

hyperid 29332
name theta232
short.name theta232

initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta233

hyperid 29333
name theta233
short.name theta233
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta234

hyperid 29334
name theta234
short.name theta234
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta235

hyperid 29335
name theta235
short.name theta235
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta236

hyperid 29336
name theta236
short.name theta236
initial 1048576
fixed FALSE
prior none
param

to.theta `function(x) x`
from.theta `function(x) x`

theta237
hyperid `29337`
name `theta237`
short.name `theta237`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta238
hyperid `29338`
name `theta238`
short.name `theta238`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta239
hyperid `29339`
name `theta239`
short.name `theta239`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta240
hyperid `29340`
name `theta240`
short.name `theta240`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta241
hyperid `29341`

name theta241
short.name theta241
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta242

hyperid 29342
name theta242
short.name theta242
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta243

hyperid 29343
name theta243
short.name theta243
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta244

hyperid 29344
name theta244
short.name theta244
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta245

hyperid 29345
name theta245
short.name theta245
initial 1048576
fixed FALSE

prior none
param
to.theta function(x) x
from.theta function(x) x

theta246
hyperid 29346
name theta246
short.name theta246
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta247
hyperid 29347
name theta247
short.name theta247
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta248
hyperid 29348
name theta248
short.name theta248
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta249
hyperid 29349
name theta249
short.name theta249
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta250

hyperid 29350
name theta250
short.name theta250
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta251

hyperid 29351
name theta251
short.name theta251
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta252

hyperid 29352
name theta252
short.name theta252
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta253

hyperid 29353
name theta253
short.name theta253
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta254

hyperid 29354
name theta254
short.name theta254

initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta255

hyperid 29355
name theta255
short.name theta255
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta256

hyperid 29356
name theta256
short.name theta256
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta257

hyperid 29357
name theta257
short.name theta257
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta258

hyperid 29358
name theta258
short.name theta258
initial 1048576
fixed FALSE
prior none
param

to.theta `function(x) x`
from.theta `function(x) x`

theta259

hyperid `29359`
name `theta259`
short.name `theta259`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta260

hyperid `29360`
name `theta260`
short.name `theta260`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta261

hyperid `29361`
name `theta261`
short.name `theta261`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta262

hyperid `29362`
name `theta262`
short.name `theta262`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta263

hyperid `29363`

name theta263
short.name theta263
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta264

hyperid 29364
name theta264
short.name theta264
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta265

hyperid 29365
name theta265
short.name theta265
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta266

hyperid 29366
name theta266
short.name theta266
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta267

hyperid 29367
name theta267
short.name theta267
initial 1048576
fixed FALSE

prior none
param
to.theta function(x) x
from.theta function(x) x

theta268
hyperid 29368
name theta268
short.name theta268
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta269
hyperid 29369
name theta269
short.name theta269
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta270
hyperid 29370
name theta270
short.name theta270
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta271
hyperid 29371
name theta271
short.name theta271
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta272

hyperid 29372
name theta272
short.name theta272
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta273

hyperid 29373
name theta273
short.name theta273
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta274

hyperid 29374
name theta274
short.name theta274
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta275

hyperid 29375
name theta275
short.name theta275
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta276

hyperid 29376
name theta276
short.name theta276

initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta277

hyperid 29377
name theta277
short.name theta277
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta278

hyperid 29378
name theta278
short.name theta278
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta279

hyperid 29379
name theta279
short.name theta279
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta280

hyperid 29380
name theta280
short.name theta280
initial 1048576
fixed FALSE
prior none
param

to.theta `function(x) x`
from.theta `function(x) x`

theta281

hyperid `29381`
name `theta281`
short.name `theta281`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta282

hyperid `29382`
name `theta282`
short.name `theta282`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta283

hyperid `29383`
name `theta283`
short.name `theta283`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta284

hyperid `29384`
name `theta284`
short.name `theta284`
initial `1048576`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta285

hyperid `29385`

name theta285
short.name theta285
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta286

hyperid 29386
name theta286
short.name theta286
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta287

hyperid 29387
name theta287
short.name theta287
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta288

hyperid 29388
name theta288
short.name theta288
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta289

hyperid 29389
name theta289
short.name theta289
initial 1048576
fixed FALSE

prior none
param
to.theta function(x) x
from.theta function(x) x

theta290
hyperid 29390
name theta290
short.name theta290
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta291
hyperid 29391
name theta291
short.name theta291
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta292
hyperid 29392
name theta292
short.name theta292
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta293
hyperid 29393
name theta293
short.name theta293
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta294

hyperid 29394
name theta294
short.name theta294
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta295

hyperid 29395
name theta295
short.name theta295
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta296

hyperid 29396
name theta296
short.name theta296
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta297

hyperid 29397
name theta297
short.name theta297
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta298

hyperid 29398
name theta298
short.name theta298

initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta299

hyperid 29399
name theta299
short.name theta299
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta300

hyperid 29400
name theta300
short.name theta300
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

constr FALSE

nrow.ncol FALSE

augmented TRUE

aug.factor 1

aug.constr 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

n.div.by -1

n.required TRUE

set.default.values TRUE

pdf iidkd

Example

Just simulate some data and estimate the parameters back. This is for `order=4`.

```
library(mvtnorm)

n <- 300
m <- 4
N <- m*n
rho <- 0.8

Sigma <- matrix(NA, m, m)
diag(Sigma) <- (1/(1:m))^2
for(i in 1:m) {
    for (j in 1:m) {
        if (i != j) {
            Sigma[i, j] <- rho^abs(i-j) * sqrt(Sigma[i, i] * Sigma[j, j])
        }
    }
}

y <- c()
yy <- rmvnorm(n, sigma = Sigma)
for(i in 1:m) {
    y <- c(y, yy[, i])
}

r <- inla(y ~ f(i, model = "iidkd", order = m, n=N,
                ## set parameters using 'theta1'.
                ## these are the default parameters.
                hyper =  list(theta1 = list(
                                  param = c(100, rep(1, m), rep(0, m*(m-1)/2))))),
          data = data.frame(i = 1:N, y),
          ## fix precision as we have exact observations
          control.family = list(hyper = list(
                                   prec = list(initial = 15, fixed = TRUE))),
          verbose = FALSE)

## this is how the internal parameters are defined
L <- t(chol(solve(Sigma)))
diag(L) <- log(diag(L))
LL <- t(chol(solve(cov(yy))))
diag(LL) <- log(diag(LL))

## compare the estimated (internal) parameters with MLE and the truth
round(dig = 3, cbind(true = c(diag(L), L[lower.tri(L)]),
                     mle = c(diag(LL), LL[lower.tri(LL)]),
                     inla = r$mode$theta))

## this gives a list of sampled matrices (stdev's and correlations)
xx <- inla.iidkd.sample(10^4, r, "i")
## compute the mean
qq <- matrix(rowMeans(matrix(unlist(xx), nrow = m^2)), m, m)

iSigma <- 1/sqrt(diag(Sigma))
Cor <- diag(iSigma) %*% Sigma %*% diag(iSigma)
round(dig = 3, cbind(inla = c(diag(qq), qq[lower.tri(qq)]),
                     true = c(sqrt(diag(Sigma)), Cor[lower.tri(Cor)])))
```

Notes

Note that for higher values of k the only integration scheme that is supported is `control.inla=list(int.strategy="eb")`.